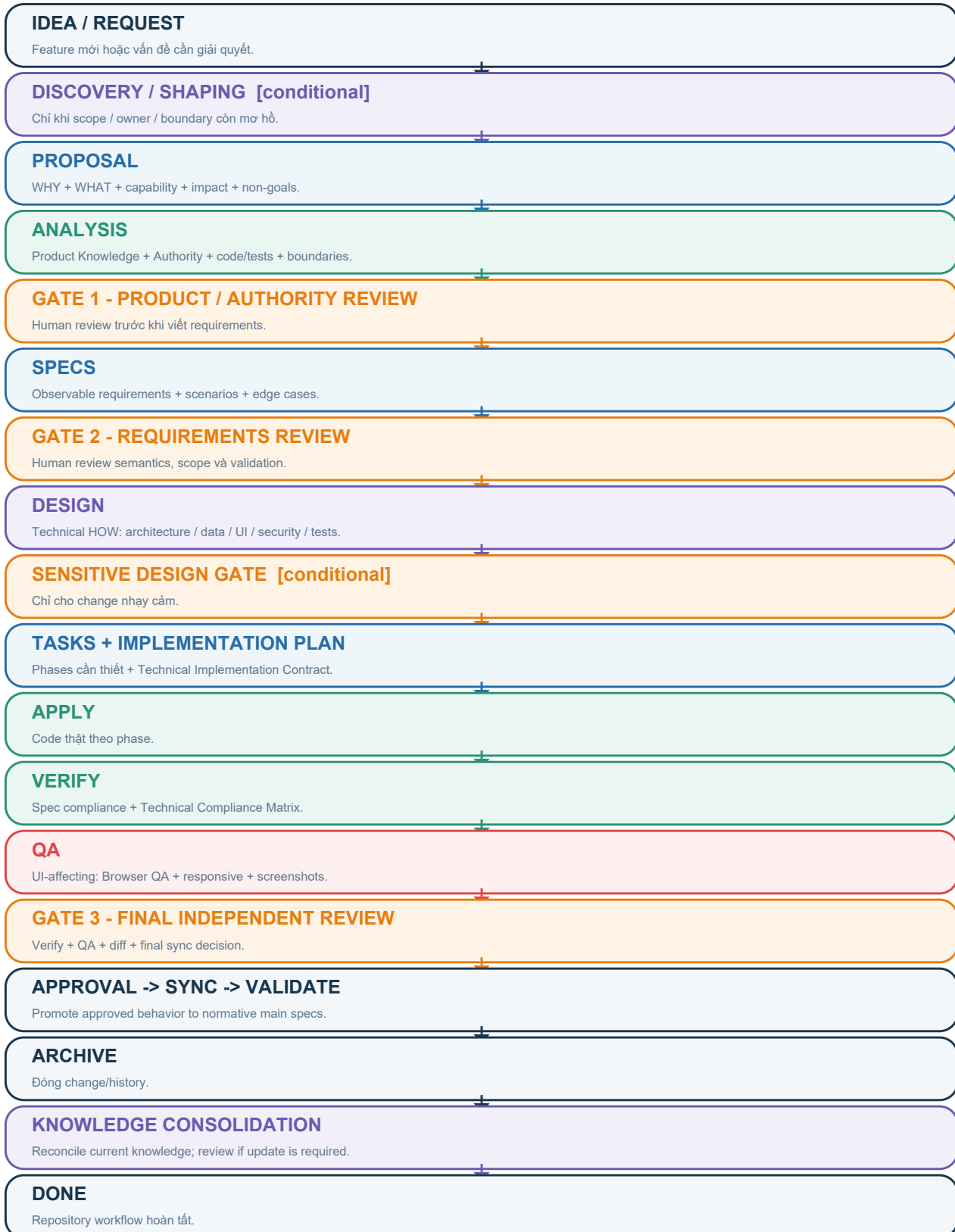


YuTa Workflow v3 - FINAL

End-to-end repository workflow (release/deploy remains a separate operational lane)



Branches và outputs quan trọng

Conditional paths được tách khỏi tuyến chính để tránh hiểu sai

NO-SPEC PATH (`skip_specs: true`)

Gate 1 approval -> không Gate 2 -> không normative spec promotion. Tiếp tục Design/Tasks/Apply/Verify/QA theo phần còn áp dụng -> Gate 3 -> Archive -> Knowledge Consolidation.

SENSITIVE DESIGN GATE [conditional]

Chỉ khi change ảnh hưởng security/authorization, data ownership/runtime, migration, payment/fiscal, legal/privacy, provider, POS transaction hoặc durable cross-module boundary.

KNOWLEDGE REVIEW [conditional]

Sau Archive: nếu Knowledge Scan = UPDATE_REQUIRED, tạo 04-knowledge-consolidation-review.md và dừng. Chỉ sau explicit approval mới apply exact approved documentation diff rồi DONE.

Review packet / output map

Proposal	proposal.md
Analysis	analysis.md
Gate 1	docs/reviews/<change>/01-analysis-review.md
Specs*	openspec/changes/<change>/specs/<capability>/spec.md
Gate 2*	docs/reviews/<change>/02-specs-review.md
Design Gate*	docs/reviews/<change>/02b-design-review.md
Tasks	tasks.md + embedded Technical Implementation Contract
QA*	docs/reviews/<change>/qa/QA_REPORT.md + screenshots
Gate 3	docs/reviews/<change>/03-final-review.md
Normative Sync*	openspec/specs/<capability>/spec.md
Knowledge Review*	docs/reviews/<change>/04-knowledge-consolidation-review.md

* conditional - Specs/Gate 2/Normative Sync are absent on the approved no-spec path; Design Gate, QA evidence, and Knowledge Review appear only when applicable.

TASKS + Technical Implementation Contract

Chỉ chọn phase change thật sự cần; phases là optional planning labels

TECHNICAL IMPLEMENTATION CONTRACT

Nằm trong Tasks/Implementation Plan, không phải OpenSpec artifact mới. Mỗi phase phải ghi boundary, canonical owner, AGENTS.md + authority sources, applicable constraints, intended files/packages, targeted checks và completion evidence.

Foundation / Data

DB/runtime owner, schema/migration consistency, tenant scope, migration/integration evidence.



Service / Domain

Trusted server scope, authorization, validation, domain purity, contract/repository ownership.



UI / Components

Server Component default, smallest Client boundary, correct component owner, reuse @yuta/ui.



Interaction / States

Mutation/form conventions, loading/error/conflict/success, keyboard/focus/accessibility.



Integration / Regression

Regression, typecheck/build, architecture/security/migration/integration, scoped diff.

APPLY completion rule

Task checkbox chỉ được tick khi implementation outcome VÀ technical compliance evidence cùng tồn tại. Nếu implementation cần phá/đổi technical boundary, STOP và quay lại Design/Product/authority gate. Không rewrite Specs/Design để khớp code.

VERIFY khác QA

Backend/DB correctness ở VERIFY; Browser QA chỉ mandatory khi UI_AFFECTING = YES

VERIFY - Technical correctness

Spec/Design -> code mapping
Tests / typecheck / build
Architecture/security
Schema/migration checks
Scoped diff
Technical Compliance Matrix

QA - User/runtime correctness

UI: Browser interaction
Responsive
Role/edit/read-only states
Loading/error/success
Accessibility + visual
Regression + screenshots

TECHNICAL COMPLIANCE MATRIX

technical rule/constraint -> authoritative source -> affected implementation -> test/check/evidence -> PASS/FAIL. VERIFY PASS chỉ khi implementation khớp Specs/Design và mọi applicable contract item PASS.

UI rule

UI_AFFECTING = YES -> BROWSER_QA_REQUIRED = YES -> Gate 3 chỉ ready khi Technical Compliance = PASS, VERIFY = PASS và QA = PASS.

Non-UI rule

Backend/DB change không cần Browser QA vô nghĩa. QA có thể là PASS theo runtime/user-facing checks phù hợp, hoặc NOT_APPLICABLE khi thật sự không có user/runtime QA dimension. Backend/DB correctness vẫn bắt buộc ở VERIFY.

QA evidence

UI QA: qa/QA_REPORT.md + screenshot-manifest.md + *.png. Default viewports khi page pack không quy định: desktop 1366x768, mobile 390x844, tablet 768x1024 khi meaningful. Status: PASS | FAIL | BLOCKED_BY_ENVIRONMENT | NOT_APPLICABLE.

\$yuta-finish-change: hai nhánh tách biệt

Branch B không chạy lại Gate 3, sync, archive hoặc implementation integrity checks

BRANCH A - Active finalization

Preconditions:

- active change tồn tại
- Gate 3 = AWAITING_HUMAN_REVIEW
- explicit final approval + sync/archive authorization
- recheck planning/diff/verify/technical-compliance hashes

Flow:

APPROVE -> SYNC -> VALIDATE -> ARCHIVE ->
KNOWLEDGE SCAN

BRANCH B - Archived Knowledge Review

Preconditions:

- Gate 3 đã APPROVED
- finish/archive đã thành công
- Workflow = AWAITING_KNOWLEDGE_REVIEW
- active change không còn
- archive tồn tại
- 04-knowledge... = AWAITING_HUMAN_REVIEW

Flow:

current-user Knowledge Review approval -> target path
set/target hashes/proposed-diff hash -> apply exact approved
documentation diff -> formatting + docs:check +
architecture:check -> record evidence -> DONE

KNOWLEDGE SCAN

Check Page/Module Product Knowledge, routing, Module Registry/lifecycle, ADR, page pack, CURRENT_STATE, resolved NEEDS REVIEW và new limitations/future work.

NO_UPDATE_REQUIRED

Ghi bounded reason + exact sources inspected ->
Workflow status DONE.

UPDATE_REQUIRED

Tạo 04-knowledge-consolidation-review.md với exact
proposed diff + target hashes. Không sửa canonical docs
trước approval.

Knowledge safety

Knowledge Review approval độc lập với Gate 3 approval. Không tự approve Product Decision, durable boundary, owner/permission, lifecycle/readiness hoặc rewrite normative specs.

Release / Deploy là operational lane riêng

Repository DONE không đồng nghĩa Production Enabled

REPOSITORY WORKFLOW DONE

Normative specs đã sync/validate (nếu có), change archived, Knowledge Consolidation đã terminal. Đây là repository completion, không phải deployment evidence.

RELEASE FOLLOW UP

Classify: NOT_REQUIRED | REQUIRED | UNKNOWN. Nếu REQUIRED, report runtime/environment, deployment/readiness evidence và post-deploy verification cần thiết.

RELEASE / DEPLOY

Authorized operational action; không tự chạy từ \$yuta-finish-change.

POST-DEPLOY VERIFY

Smoke/health/runtime checks trên environment thực.

MONITOR / ROLLBACK

Observability, incident/recovery evidence khi applicable.

Nguyên tắc cuối

IMPLEMENTED != PRODUCTION_ENABLED | VERIFY != QA | QA PASS != Production Readiness
Archive != Knowledge current | Sync != Product approval | Workflow progress != lifecycle promotion